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Md. Masum Billah

Full-Stack & AI Engineer | Mobile, Web Intelligent Systems | Data Science



I am a **Computer Science & Engineering graduate** and a dedicated **problem solver** with a strong track record on **Codeforces** and **LeetCode**. I have hands-on experience in **Full-Stack Development** and **Machine Learning**, with **15+ real-world projects** including mobile apps, web platforms, and intelligent systems.

I specialize in building **real-world, data-driven solutions** by creating **custom datasets**, developing **ML models**, and integrating **REST APIs** for practical applications such as environmental monitoring and smart systems. With experience in **Flutter**, backend development, and **Linux-based deployment**, I bridge the gap between **AI research** and **production-ready software**.

My objective is to work as a **Software Engineer** in a **growth-oriented organization** where I can build **scalable and impactful applications** while continuously enhancing my skills in **mobile app development**, **web technologies**, and **artificial intelligence**.

SKILLS

Languages	Dart, Python, C++, Java, PHP, JavaScript
Frontend & Mobile	Flutter, HTML5, CSS3, Streamlit, Responsive UI/UX
Backend & Cloud	Django, RESTful APIs, MySQL/PostgreSQL, Firebase, IoT Systems.
AI & Machine Learning	Computer Vision, YOLO, PyTorch, Scikit-learn, Hybrid ML Models, Model Evaluation, AI Automation (n8n)
DevOps & Tools	Git, GitHub, Docker, Ubuntu/Linux, Windows
Core Competencies	Data Structures & Algorithms, Competitive Programming (Codeforces, LeetCode)

PROFESSIONAL EXPERIENCE

Executive Engineer, AI & Mobile App <i>Softvance Agency Mohakhali, Dhaka, Bangladesh</i>	Feb 2026 – Present
- Developed and maintained cross-platform mobile applications using Flutter for Android and iOS. - Built responsive and user-friendly interfaces following modern UI/UX design principles. - Integrated REST APIs and third-party services to enable seamless communication between applications and backend systems. - Designed and developed scalable REST APIs and backend services for web and mobile applications. - Built full-stack web applications using modern frontend and backend technologies. - Integrated AI-powered features such as chatbots, recommendation systems, and workflow automation into software applications.	

SELECTED PROJECTS

Hybrid ML-Based Real-Time Road Sign & Vehicle Detection System <i>Computer Vision, Deep Learning</i>	2024–2025
- I created a custom Dataset annotated with 9,133 images and 29 classes of Bangladeshi road signs. - Implemented a YOLO/Hybrid ML pipeline for real-time sign and vehicle detection. - Developed a Streamlit-based web application supporting image, video, and live camera inference. - Optimised performance for crowded and low-light road environments.	
Air Quality Index Monitoring Platform <i>Google Earth Engine(GEE), Weather API, Data Analysis</i>	2025
- Built an environmental monitoring system using Google Earth Engine (GEE). - Processed satellite data to compute and visualize Air Quality Index (AQI) trends. - Integrated REST APIs to fetch real-time environmental data. - Designed dashboards for spatial and temporal analysis of pollution levels.	
IoT-Based Health Monitoring Band With MobileApp <i>Flutter, Firebase, Embedded Systems</i>	2024
- Built a wearable health-monitoring band using ESP8266, MAX30100, GPS, and temperature sensors. - Developed a Flutter mobile application for real-time data visualization and alerts. - Integrated Firebase for live cloud synchronization and remote monitoring.	
Stroke Prediction App – Streamlit Machine Learning <i>Python, Streamlit, Scikit-learn, Pandas, Data Analysis</i>	2024
- Developed an end-to-end stroke prediction web application using Streamlit for interactive deployment. - Performed data cleaning, preprocessing, encoding, and exploratory data analysis on a self-curated healthcare dataset. - Implemented and compared Machine Learning models including Logistic Regression and Random Forest for classification. - Designed an intuitive UI for real-time stroke risk prediction based on user health inputs.	

FoodHaven Flutter-Based Food Ordering App <i>Flutter, UI/UX, State Management</i>	2024
• Developed a complete Flutter app for food ordering and delivery. • Implemented product listing, cart, OTP authentication, and responsive UI. • Integrated backend services for order processing and user profiles.	
Six Seasons – OpenGL Seasonal Animation <i>OpenGL, Computer Graphics, C++</i>	2023
• Designed a graphical simulation showing six seasonal environments using OpenGL. • Implemented animated elements such as weather, rain, snowfall, and lighting transitions. • Optimized rendering performance and scene composition for smooth visualization.	

EDUCATION

Bachelor of Science in Computer Science & Engineering <i>Bangladesh University of Business and Technology (BUBT)</i>	Jan 2022 – Dec 2025
CGPA: 3.78/4.00	
<i>Relevant Coursework:</i> Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Computer Networks, Machine Learning, Artificial Intelligence, Neural Networks and Fuzzy Systems, Pattern Recognition, Knowledge Engineering, Operating Systems, Cyber Security & Forensic Digital, Internet of Things(IoT), Computer Network.	

COURSES AND CERTIFICATIONS

• n8n Automation: Agentic AI, Workflows & GenAI - Udemy	2026
• Machine Learning A–Z™: AI, Python & R + ChatGPT Bonus - Udemy	2025
• The Complete JavaScript Course: From Zero to Expert - Udemy	2025
• Flutter & Dart: The Complete Guide - Udemy	2024
• Mastering Java SE Programming: From Beginner to Master - Coursera	2023
• Beginner’s Guide to Competitive Programming using C++ - Udemy	2022

ACTIVITIES

Contestant, ICPC Asia Dhaka Regional Contest - BUBT (Preliminary Round)	2023
Contestant, BUBT Intra University Programming Contest	2023
Volunteering, ICPC Asia Dhaka Regional Contest - BUBT [Main Event]	2023
ICT Lead, ESAB BUBT Unit	Fall 2023 - Spring 2024
Strategic Member, BUBT IT Club	2024
First Place, <i>Brainstorming</i> - Project Showcase	Spring 2024
First Place, <i>Brainstorming</i> - Poster Presentation on Machine Learning	Spring 2024
First Place, <i>Brainstorming</i> - Research Paper Presentation	Fall 2025
Volunteering, Technical Team Lead, ICPC Asia Dhaka Regional Contest - BUBT [Main Event]	2025

REFERENCES

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